

SOP-HVAC-3700: HVAC System Maintenance and Control Room Pressurization

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Section 1. Purpose and Scope

Governs preventive maintenance of HVAC systems, with particular emphasis on control room/building pressurization systems that protect occupants from external hazardous atmosphere.

Section 2. Pressurization Verification

Clause 2.1 — Positive-pressure control buildings shall maintain at least 0.05 inches water column positive pressure relative to outside; pressurization is verified continuously by a monitored differential pressure gauge with a low-pressure alarm.

Clause 2.2 — A pressurization alarm requires immediate investigation; occupants shall not be directed to evacuate the building on a pressurization alarm alone unless combined with a confirmed external gas release — the building may still offer better protection than the outside atmosphere, per the site's building protection philosophy.

Section 3. Filter Maintenance

Clause 3.1 — Intake filters (including any charcoal/gas-phase filtration on a hazard-protective building) are inspected monthly and replaced per the manufacturer's loading schedule or sooner if differential pressure across the filter indicates loading.

Section 4. General HVAC PM

Clause 4.1 — General HVAC preventive maintenance (belt inspection, refrigerant charge check, coil cleaning) is performed per the equipment-specific PM schedule, at minimum semi-annually.

Section 5. Records

Clause 5.1 — Pressurization monitoring logs and PM records are filed to the Facilities File and retained for 3 years.